DAX for week 5

Data Science & Analytics Internship · the measures you need and nothing else · type them, do not read them

DAX is the language Power BI uses for calculations. It looks like Excel formulas and behaves quite differently, which is why people who are good at Excel often find it frustrating at first.

This covers the eight things your week 5 measures need. Write each one as you read it — a measure you have typed is worth ten you have skimmed.

1 — Measure or calculated column?

The first thing to get right, and the thing beginners get wrong most often. They look almost identical and do completely different jobs.

	Calculated column	Measure
Calculated	Once, when the data loads. Row by row.	Every time you look at it, for whatever is on screen.
Stored	In the table. Takes up memory.	Nowhere. Worked out on demand.
Use it for	Something that belongs to one row — a category band, a flag, a year extracted from a date.	Anything you aggregate — totals, averages, counts, rates, percentages.
Made with	Modeling → New column	Modeling → New measure

If in doubt, make a measure. Nearly everything you need this week is a measure. A calculated column that sums a value is almost always a mistake — it gives you the same number on every row and bloats the file.

2 — Your first measures

Modeling → **New measure**, then type into the formula bar. Put them on your fact table.

```
Total Revenue = SUM(sales_clean[line_total])

Transactions = COUNTROWS(sales_clean)

Customers = DISTINCTCOUNT(sales_clean[customer_id])

Average Sale = AVERAGE(sales_clean[line_total])
```

Function	Gives you	Watch out for
`SUM`	Adds a column up	Numbers only.
`COUNTROWS`	How many rows	Takes a table, not a column.
`COUNT`	Non-blank values in a column	Skips blanks — so it may be lower than COUNTROWS.
`DISTINCTCOUNT`	How many different values	Slow on very large columns. Fine on yours.
`AVERAGE`	Mean	Ignores blanks. That may or may not be what you want.
`MIN / MAX`	Smallest / largest	Works on dates too.
`MEDIAN`	Middle value	Better than AVERAGE when the data is skewed.

3 — Always use DIVIDE

```
-- wrong: blows up on a zero
Bad Rate = [Defaults] / [Applications]

-- right
Completion Rate = DIVIDE([Completed Trips], [Total Trips])

-- right, with something to show instead of blank
Completion Rate = DIVIDE([Completed Trips], [Total Trips], 0)
```

DIVIDE handles division by zero. The **/** operator throws an error that appears in your dashboard as an ugly box. There is no reason ever to use **/** between two measures.

4 — Filter context: why measures change

This is the central idea in DAX, and it is genuinely different from Excel.

A measure has no fixed value. It is recalculated for whatever is on screen at that moment. Drop **Total Revenue** into a card and you get the grand total. Put it in a table beside **Category** and each row recalculates for just that category. Add a slicer for 2025 and every number changes again.

```
Total Revenue = SUM(sales_clean[line_total])

    in a card                      5,392,644
    beside Category = Beverages    1,204,118
    beside Category = Beverages
       with a 2025 slicer            612,440

one measure, three answers — the filters decide
```

You never write the filters into the measure. The visual applies them. That is why one well-written measure serves your whole report.

5 — CALCULATE: changing the filters

The most important function in DAX. It takes a calculation and applies your own filters to it.

```
CALCULATE( <the calculation>, <filter>, <filter>, ... )
```

```
Mobile Money Revenue =
CALCULATE(
    [Total Revenue],
    sales_clean[payment_method] = "Mobile Money"
)

Completed Trips =
CALCULATE(
    COUNTROWS(trips_clean),
    trips_clean[status] = "Completed"
)
```

Note that the first measure reuses **[Total Revenue]** rather than repeating the SUM. Build measures out of other measures — when the definition changes, you fix it in one place.

Percentage of total

ALL removes filters, which is how you get a denominator that ignores what the visual is filtered to.

```
% of Total Revenue =
DIVIDE(
    [Total Revenue],
    CALCULATE([Total Revenue], ALL(products_clean))
)
```

*In a table by category, the numerator is that category and the denominator is every category. Without **ALL** both would be filtered the same way and every row would read 100%.*

6 — Time intelligence

These only work if you built a date table and **marked it as a date table** in week 4. If they return blank, that is the first thing to check.

```
Revenue LY =
CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(dim_date[Date]))

Revenue YoY % =
DIVIDE([Total Revenue] - [Revenue LY], [Revenue LY])

Revenue YTD =
TOTALYTD([Total Revenue], dim_date[Date])

Revenue Prev Month =
CALCULATE([Total Revenue], DATEADD(dim_date[Date], -1, MONTH))
```

Blank results are almost always the date table. Either it is not marked as a date table, or it does not cover your full date range, or the relationship to your fact table is missing. Check those three before rewriting the measure.

7 — Writing it readably

You will be reading these in week 8 when someone asks how a number was calculated.

```
-- hard to read
Rev=CALCULATE(SUM(sales_clean[line_total]),sales_clean[payment_method]="Cash")

-- easy to read
Cash Revenue =
CALCULATE(
    [Total Revenue],
    sales_clean[payment_method] = "Cash"
)
```

- **Name it in plain English.** *Cash Revenue*, not *Rev2*. The name is what your stakeholder sees on the chart.
- **Shift+Enter** gives you a new line in the formula bar. Use it.
- **Two dashes start a comment.** One line saying why is worth a lot in week 8.
- **Format the measure** once it works — Measure tools → Format → Currency or Percentage, and set decimal places. Nobody wants to read 5392644.3847261.

8 — Checking a measure is right

A measure that returns a plausible wrong number is worse than one that errors, because nobody notices. Check every one against SQL.

```
-- in DBeaver
SELECT round(sum(line_total), 2) FROM group1a.sales_clean;

-- then drop [Total Revenue] into a card in Power BI
-- the two numbers must match exactly
```

If they do not match, the measure is not usually the problem. Look at the model first — a relationship with the wrong cardinality or direction silently drops or duplicates rows.

The measures you need this week

Five is plenty. Each should answer part of your question.

Domain	Suggested measures
Retail	Total Revenue · Transactions · Average Sale · Units Sold · Revenue YoY %
Clinic	Average Wait · Median Wait · Visits · % Emergency Visits · Fee Revenue
School	Average Score · Pass Rate · Average Attendance · Results Recorded · Term-on-Term Change
Rides	Total Fares · Trips · Completion Rate · Average Fare · Cancellations

Two worth writing carefully

```
Pass Rate =
DIVIDE(
    CALCULATE(COUNTROWS(results_clean), results_clean[score] >= 50),
    COUNTROWS(results_clean)
)

Completion Rate =
DIVIDE([Completed Trips], COUNTROWS(trips_clean))
```

A pass mark of 50 is an assumption. Write it down in your week 5 form and be ready to defend it — the same kind of decision as dayfirst in week 3.

Errors you will see

Message	What it means
`Cannot find table or column`	Spelling, or you renamed the column in week 4 and the measure still uses the old name. Start typing and let autocomplete fill it in.
`A single value for column cannot be determined`	You used a column where an aggregation was needed. Wrap it in SUM or AVERAGE.
`The measure returns blank everywhere`	For time intelligence: the date table. Otherwise a filter that matches nothing — check your spelling against the actual values.
`Numbers are far too large`	A many-to-many relationship duplicating rows. Fix the model, not the measure.
`Every row shows the same number`	A missing relationship, so the filter never reaches your fact table. Check Model view.
`Infinity, or a division error`	You used / instead of DIVIDE.

Your first dashboard

Once the measures work, build one page. **Ugly is expected this week.** You are proving the numbers are right, not designing anything.

A layout that works

- **Cards along the top** for your headline measures. Three or four, no more.
- **A line chart** of your main measure against **dim_date**, not the fact table's own date column. That is the point of the date table.
- **A bar chart** broken down by your main dimension.
- **Slicers at the top** for date and one category.

Check the totals before you go any further. The number on your card must match the SQL. If it does not, stop and fix the model — every chart you build on top of it will be wrong in the same way, and you will not find out until week 7.

Before you finish week 5

- Five measures written, named in plain English, formatted.
- Every measure checked against a SQL query.
- One dashboard page with cards, a trend and a breakdown.
- At least one slicer, tested — click it and watch every visual change.
- Your assumptions written down. What counts as a pass? What counts as completed?
- The **.pbix** committed to **dashboard/**.

The test for this week is not whether it looks good. It is whether a number on your dashboard and the same number from SQL agree. Get that right and week 6 is design work. Get it wrong and week 6 is archaeology.